A proof of Seymour's second-neighborhood conjecture

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Abstract

We prove that every nonempty finite oriented graph has a vertex with at least as many vertices at directed distance two as at directed distance one. This resolves Seymour's second neighborhood conjecture positively. The argument is presented as a sequence of individually justified steps.

1 The result and the plan

Statement

An *oriented graph* is a finite directed graph with no loops, no multiple arcs, and no pair of opposite arcs; its underlying undirected graph need not be complete or connected. We write $x \to y$ for an arc. For a vertex v,

$$N_1^+(v) = \{u : v \to u\}, \qquad N_2^+(v) = \{w \notin \{v\} \cup N_1^+(v) : v \to u \to w \text{ for some } u\}.$$

Thus $N_2^+(v)$ is the set of vertices at directed distance exactly two from v, each counted once.

Theorem 1.1. *Every nonempty finite oriented graph has a vertex v such that $|N_1^+(v)| \le |N_2^+(v)|$.*

There is no connectivity or degree hypothesis; a sink ($N_1^+(v) = \varnothing$) satisfies the conclusion trivially. In a directed triangle every vertex has $|N_1^+| = |N_2^+| = 1$, so the inequality cannot be made strict.

Image operator. For a digraph D and $S \subseteq V(D)$ put

$$F_D(S) = \{y : x \to y \text{ for some } x \in S\}, \qquad F_D^2(S) = F_D(F_D(S)).$$

The set $F_D(S)$ may meet S. The operator is monotone and $F_D(S \cup S') = F_D(S) \cup F_D(S')$. In an oriented graph no walk of length two returns to its start, so $v \notin F_D^2(\{v\})$ and $N_2^+(v) = F_D^2(\{v\}) \setminus F_D(\{v\})$.

Context

The conjecture was recorded by Dean and Latka [8]. For tournaments (Dean's conjecture) it was proved by Fisher [12] using Farkas' lemma and by Havet and Thomassé [14] using median orders; extensions to tournaments missing small structures appear in [11, 13, 7, 6]. Small minimum outdegree is treated in [16, 18], approximate ratios in [5, 15], orientations of random graphs in [2, 10], and a dense case in [4]. The present proof builds on the minimal-counterexample and graph-product methods of Brantner, Brockman, Kay and Snively [3] and on Seacrest's subset perspective [20]; the main additional ingredient is a pruning bound for arbitrary finite binary relations (Lemma 3.1). Theorem 1.1 also gives the weighted forms of the conjecture through Seacrest's equivalences [19] (Section 5) and consequences for short directed cycles (Section 6).

Proof at a glance

The theorem is proved by contradiction, in three parts.

(A) *Reduction* (Section 2, Proposition 2.1). Take a counterexample D of minimum order and then minimum number of arcs. Deleting arcs and invoking minimality shows $|F_D^2(S) \setminus F_D(S)| < |F_D(S) \setminus S|$ for every nonempty proper S (Lemma 2.2). Replacing each vertex by a large transitive tournament (Lemma 2.3) yields an oriented graph G with positive indegrees and

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad (\varnothing \neq U \subsetneq V(G)). \tag{1}$$

(B) *Pruning* (Section 3, Lemma 3.1); independent of (A) and of orientation. Two families of ordered pairs over a set with a binary relation may "conflict". All conflicts can be removed by deleting members, at a cost bounded by the size of a set H plus the number of certain points left uncovered *after* deletion. Deletions correspond to vertex covers of an auxiliary bipartite graph. By Kőnig's argument it suffices to bound a maximum matching, and this is done by a rank computation with four matrices satisfying $NL = AB$ at generic coefficients (Lemma 3.2 is the linear-algebra input).

(C) *Contradiction* (Section 4, Proposition 4.1): no G as in (A) exists. With $X = V(G)$, consider pairs (P, Q) of subsets of $X \times X$ containing the diagonal such that F_1P (advance the first coordinate along an arc) and F_2Q (advance the second) are disjoint. Let $M = |P| + |Q|$, let d be the number of points in neither image, and maximize $M + d$. Applying (1) to the columns of P and the rows of Q shows that the complements of F_2^2Q and F_1^2P have combined size $> M + 2d$. They may conflict; pruning deletes at most $d + d'$ members, where d' counts the points uncovered afterwards, and keeps the diagonal. The new objective exceeds $M + 2d - (d + d') + d' = M + d$, a contradiction. The cancellation of d' is why the pruning cost must be charged to points uncovered after deletion.

Dependencies: Lemma 2.2 $\Rightarrow$ Lemma 2.3 $\Rightarrow$ Proposition 2.1; Lemma 3.2 $\Rightarrow$ Lemma 3.1 $\Rightarrow$ Proposition 4.1; the two propositions give Theorem 1.1, which gives Corollaries 5.1 and 6.1.

Renamed symbols. To keep one meaning per symbol: in Lemma 2.2 the lost out-neighbors form Y (source: Q) and the degrees are ν_1, ν_2 (source: d_1, d_2); in Lemma 2.3 label sets are U_x (source: T_v); in Section 3 the support-rank matrices are $\Theta, \Theta^{p,i}$ with supports $\Omega, \Omega^{p,i}$ (source: D, T) and vectors are bold; in Section 5 we write $\mathrm{Arc}(D)$, ω, γ_v^ω, Φ_D (source: $A(D)$, w, β_v^w, M_D); in Section 6 we write $\overline{N}(v)$, $\deg^\pm$ (source: $Z(v)$, $d^\pm$).

2 From a counterexample to a strict subset inequality

In this section D is a base oriented graph and $F = F_D$.

Proposition 2.1 (Counterexample reduction)**.** *If Theorem 1.1 has a counterexample, then there is a finite nonempty oriented graph G in which every vertex has positive indegree and*

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \textit{for every } \varnothing \neq U \subsetneq V(G). \tag{2}$$

The proof, at the end of this section, combines Lemma 2.2 (minimality gives a strict deficit for subsets) with Lemma 2.3 (a blow-up amplifies it). The proposition is used only in the proof of Theorem 1.1.

The arc-deletion argument adapts the proof idea of [20, Lemma 4]. With Seacrest's parameter λ equal to 1, his statement gives $|F^2(S) \setminus F(S)| < |F(S)|$ in an edge-minimal counterexample when $F(S) \neq \varnothing$. Lemma 2.2 replaces the right-hand side by $|F(S) \setminus S|$ for nonempty proper S, under the stronger minimality below; it is proved in full.

Lemma 2.2 (Subset deficit)**.** *Let D be a finite nonempty oriented graph such that $|N_1^+(v)| > |N_2^+(v)|$ for every $v \in V(D)$. Choose D with minimum order and, subject to that, minimum number of arcs among all graphs with this property. Then D is strongly connected, every vertex has positive indegree, and*

$$|F^2(S) \setminus F(S)| < |F(S) \setminus S| \qquad \text{for every } \varnothing \neq S \subsetneq V(D). \tag{3}$$

Proof. Plan. After two structural steps we fix S and grow a set T from $\varnothing$ to $E = F(S) \setminus S$. Each growth step comes from deleting arcs and invoking arc-minimality, and it increases a counting function g by strictly less than the number of vertices added. Throughout, $N_i(x)$ means $N_i^+(x)$ computed in D.

Step 1 (no sink). A sink x has $|N_1(x)| = 0 \leq |N_2(x)|$, against the hypothesis. So D has no sink, hence at least two vertices.

Step 2 (strong connectivity). Suppose D is not strongly connected. Then some strong component $V_0 \neq V(D)$ has no arc leaving it (a terminal component of the acyclic condensation). For $x \in V_0$ every walk from x stays in V_0, so $N_1(x)$ and $N_2(x)$ are the same in the induced subgraph $D[V_0]$ as in D. Thus $D[V_0]$ satisfies the hypothesis and has fewer vertices, contradicting minimum order. So D is strongly connected; having at least two vertices, each vertex has an in-neighbor.

Step 3 (set-up). Fix $\varnothing \neq S \subsetneq V(D)$ and put

$$I = F(S) \cap S, \qquad E = F(S) \setminus S, \qquad g(T) = |F(T) \setminus (I \cup T)| \quad (T \subseteq E).$$

So $F(S) = I \sqcup E$. Strong connectivity gives an arc from S to its complement, so $E \neq \varnothing$.

Step 4 (delete arcs). Fix $T \subsetneq E$. Let D_T be D with all arcs from S to $E \setminus T$ deleted; write N_i', F' for neighborhoods and images in D_T. Every vertex of $E \setminus T \neq \varnothing$ is the head of an arc from S, so D_T is an oriented graph of the same order with fewer arcs. By the choice of D it fails the hypothesis: some vertex v has

$$|N_1'(v)| \leq |N_2'(v)|. \tag{4}$$

Step 5 (v lost out-neighbors). Let $Y = N_1(v) \setminus N_1'(v)$. If $Y = \varnothing$, then $N_1'(v) = N_1(v)$ and $N_2'(v) \subseteq N_2(v)$ (a two-step path in D_T is one in D, and the excluded set $\{v\} \cup N_1(v)$ is unchanged), so $|N_1(v)| \leq |N_2'(v)| \leq |N_2(v)|$, against the hypothesis on D. Hence $Y \neq \varnothing$. Only arcs from S into $E \setminus T$ were deleted, so $v \in S$ and $Y = N_1(v) \cap (E \setminus T)$. Since $N_1(v) \subseteq F(S) = I \cup T \cup (E \setminus T)$,

$$N_1(v) \subseteq I \cup T \cup Y, \qquad N_1'(v) = N_1(v) \setminus Y \subseteq I \cup T, \qquad Y \cap (I \cup T) = \varnothing. \tag{5}$$

Step 6 (two-step endpoints in D_T). We claim

$$F'^2(\{v\}) \subseteq I \cup T \cup F(T). \tag{6}$$

Let $v \to u \to w$ in D_T. By (5), $u \in I \cup T$. If $u \in I \subseteq S$, then $w \in F(S) = I \cup E$, and $w \notin E \setminus T$ because the arc $u \to w$ survived; so $w \in I \cup T$. If $u \in T$, then $w \in F(T)$.

Step 7 (gained second out-neighbors). $N_2'(v) \setminus N_2(v) \subseteq Y \cap F(T)$. Indeed, let w lie in the left side. Then $w \neq v$ and w ends a two-step path from v in D_T, hence in D; as $w \notin N_2(v)$, necessarily $w \in N_1(v)$. As $w \in N_2'(v)$, we have $w \notin N_1'(v)$. So $w \in Y$. By (6) and $Y \cap (I \cup T) = \varnothing$, $w \in F(T)$.

Step 8 (lost second out-neighbors). Put

$$C = F(Y) \setminus (I \cup T \cup Y \cup F(T)).$$

Then $C \subseteq N_2(v) \setminus N_2'(v)$. Indeed, let $w \in C$ and pick $y \in Y$ with $v \to y \to w$ in D. By (5), $w \notin N_1(v)$; and $w \neq v$, since $y \to v$ would oppose $v \to y$. So $w \in N_2(v)$. By (6), $w \notin F'^2(\{v\}) \supseteq N_2'(v)$.

Step 9 (count). Steps 7 and 8 give $N_2'(v) \subseteq (N_2(v) \setminus C) \cup (Y \cap F(T))$ with $C \subseteq N_2(v)$. Write $\nu_i = |N_i(v)|$, so $\nu_2 < \nu_1$. Then (4) and (5) yield

$$\nu_1 - |Y| = |N_1'(v)| \leq |N_2'(v)| \leq \nu_2 - |C| + |Y \cap F(T)|. \tag{7}$$

Step 10 (increment of g). Since $F(T \cup Y) = F(T) \cup F(Y)$,

$$F(T \cup Y) \setminus (I \cup T \cup Y) = \Big[\big(F(T) \setminus (I \cup T)\big) \setminus \big(Y \cap F(T)\big)\Big] \sqcup C,$$

the bracket being the part inside $F(T)$ and C the part outside. As $Y \cap F(T) \subseteq F(T) \setminus (I \cup T)$ by (5), taking cardinalities and using (7) gives

$$g(T \cup Y) - g(T) = |C| - |Y \cap F(T)| \leq |Y| + \nu_2 - \nu_1 < |Y|. \tag{8}$$

Step 11 (iterate). Set $T_0 = \varnothing$. While $T_k \neq E$, apply Steps 4–10 with $T = T_k$ (always deleting from the original D, so minimality applies each time) to get a nonempty $Y_k \subseteq E \setminus T_k$, and put $T_{k+1} = T_k \sqcup Y_k$. The sets increase strictly inside the finite set E, so $T_r = E$ for some $r \geq 1$. Summing (8) and using $g(\varnothing) = |F(\varnothing) \setminus I| = 0$ gives $g(E) < \sum_k |Y_k| = |E|$.

Step 12 (conclusion). Since $F(S) = I \cup E$ and $F(I) \subseteq F(S)$ (as $I \subseteq S$),

$$F^2(S) \setminus F(S) = \big(F(I) \cup F(E)\big) \setminus (I \cup E) = F(E) \setminus (I \cup E),$$

whose cardinality is $g(E) < |E| = |F(S) \setminus S|$. □

The next lemma amplifies the integer deficit in (3). Its construction is the lexicographic product of D with a transitive tournament, as in [3, Theorem 4.3]; here an estimate is needed for every nonempty proper subset, not only for single vertices. It is used only in the proof of Proposition 2.1.

Lemma 2.3 (Transitive-tournament blowup)**.** *Let D be an oriented graph on $n \geq 1$ vertices, all of positive indegree, satisfying* (3) *for every nonempty proper subset of $V(D)$. For an integer $m > n$, replace each vertex by a transitive tournament on m vertices. For every arc $v \to w$ of D, put all arcs from the block at v to the block at w, and put no other arcs between distinct blocks. The resulting graph G is oriented, all its indegrees are positive, and it satisfies* (2)*.*

Proof. Plan. Split U by blocks. One unit of base-level deficit is worth a whole block (m vertices), while effects inside blocks cost at most one per base vertex, at most $n < m$ in total.

Step 1 (inside a block). Label each block by $1, \ldots, m$ with arcs from smaller to larger labels, and let f be the image operator of this tournament on label sets. For a nonempty label set Λ we have $f(\Lambda) = \{x : x > \min \Lambda\}$, so $\Lambda \subseteq \{\min \Lambda\} \cup f(\Lambda)$ and $f^2(\Lambda) \subseteq f(\Lambda)$. Hence

$$|\Lambda| - |f(\Lambda)| \leq 1, \qquad |\Lambda| - 2|f(\Lambda)| + |f^2(\Lambda)| \leq |\Lambda| - |f(\Lambda)| \leq 1. \tag{9}$$

Step 2 (G is oriented with positive indegrees). Call an arc of G *internal* if it lies in one block and *external* otherwise. Internal arcs form transitive tournaments; between two blocks all arcs point one way, because D has no opposite arcs. So G is oriented. A vertex in the block at x receives arcs from the whole block at any in-neighbor of x.

Step 3 (decomposition). Fix $\varnothing \neq U \subsetneq V(G)$. Let U_x be the set of labels of U in the block at x, and

$$S = \{x \in V(D) : U_x \neq \varnothing\} \neq \varnothing, \qquad H_0 = F(S) \cup F^2(S).$$

Thus $|U| = \sum_{x \in S} |U_x|$.

Step 4 (first image). If $y \in F(S)$, some $x \in S$ has $x \to y$, and a vertex of U in the block at x sends arcs to the whole block at y. If $y \notin F(S)$, no external arc from U enters the block at y, so $F_G(U)$ meets it exactly in $f(U_y)$ (empty if $y \notin S$). Hence

$$|F_G(U)| = m|F(S)| + \sum_{x \in S \setminus F(S)} |f(U_x)|.$$

Step 5 (second image). Consider a two-step path starting at a vertex of U in the block at $x \in S$. With two external arcs it ends in a block of $F^2(S)$; with exactly one external arc, in either order, in a block of $F(S)$; with none, at a label of $f^2(U_x)$ in the block at x. Bounding each block of H_0 by m,

$$|F_G^2(U)| \leq m|H_0| + \sum_{x \in S \setminus H_0} |f^2(U_x)|.$$

Step 6 (combine). Steps 3–5 give

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| \leq m\big(|H_0| - 2|F(S)|\big) + \sum_{x \in S} |U_x| \\ - 2 \sum_{x \in S \setminus F(S)} |f(U_x)| + \sum_{x \in S \setminus H_0} |f^2(U_x)|. \end{aligned} \tag{10}$$

Since $F(S) \subseteq H_0$, the last sum may be extended to $S \setminus F(S)$; this only adds nonnegative terms. Now bound $|U_x| \leq m$ for $x \in S \cap F(S)$, and apply (9) with $\Lambda = U_x$ for each $x \in S \setminus F(S)$. The right side of (10) is then at most

$$m\big(|H_0| - 2|F(S)| + |S \cap F(S)|\big) + |S \setminus F(S)|.$$

Using $|H_0| = |F(S)| + |F^2(S) \setminus F(S)|$, $|F(S)| - |S \cap F(S)| = |F(S) \setminus S|$ and $|S \setminus F(S)| \leq n$,

$$|U| - 2|F_G(U)| + |F_G^2(U)| \leq m\big(|F^2(S) \setminus F(S)| - |F(S) \setminus S|\big) + n. \tag{11}$$

Step 7 (case $S \subsetneq V(D)$). By (3) the bracket in (11) is an integer ≤ -1, so the right side is at most $-m + n < 0$.

Step 8 (case $S = V(D)$). Here (3) is unavailable. Positive indegrees in D give $F(S) = V(D)$, so $F_G(U) = V(G)$ by Step 4; positive indegrees in G give $F_G^2(U) = F_G(V(G)) = V(G)$. Hence $|U| - 2|F_G(U)| + |F_G^2(U)| = |U| - |V(G)| < 0$, as U is proper. □

Proof of Proposition 2.1. Among counterexamples choose D of minimum order and then minimum number of arcs. Lemma 2.2 supplies positive indegrees and (3); Lemma 2.3 with any integer $m > |V(D)|$ supplies G. □

3 Pruning two families of pairs

This section is self-contained: X is any finite set with any binary relation. Lemma 3.1 is used once, in Step 9 of the proof of Proposition 4.1.

Lemma 3.1 (Pruning)**.** *Let X be a finite set with a binary relation denoted by $\to$, and let $\mathcal{R}, \mathcal{C} \subseteq X \times X$ be two families, regarded as separate copies even when their coordinates agree. Say that $(p, j) \in \mathcal{R}$ and $(i, s) \in \mathcal{C}$* conflict *if $p \to i$ and $s \to j$. Define*

$$Z = \{(i, j) : \textit{some } (p, j) \in \mathcal{R} \textit{ conflicts with some } (i, s) \in \mathcal{C}\},$$
$$H = \{(p, s) : \textit{some } (p, j) \in \mathcal{R} \textit{ conflicts with some } (i, s) \in \mathcal{C}\}.$$

A point (i, j) is covered *by a member $(p, j) \in \mathcal{R}$ when $p \to i$, and by a member $(i, s) \in \mathcal{C}$ when $s \to j$; these are the only ways a member covers a point.*

There are subfamilies $\mathcal{R}' \subseteq \mathcal{R}$ and $\mathcal{C}' \subseteq \mathcal{C}$ with no conflicts between them such that every initially conflict-free member is retained and

$$|\mathcal{R} \setminus \mathcal{R}'| + |\mathcal{C} \setminus \mathcal{C}'| \le |H| + |\{z \in Z : z \textit{ is covered by no member of } \mathcal{R}' \cup \mathcal{C}'\}|. \tag{12}$$

Deleted members of $\mathcal{R}$ and of $\mathcal{C}$ are counted separately.

Reading the definitions. A member $(p, j) \in \mathcal{R}$ covers the points obtained by advancing its first coordinate along $\to$; a member $(i, s) \in \mathcal{C}$ covers those obtained by advancing its second coordinate. Two members conflict exactly when they cover a common point (i, j); Z is the set of such points and H collects the opposite corners (p, s). *Example:* $X = \{1, 2\}$ with only $1 \to 2$, $\mathcal{R} = \{(1, 2)\}$, $\mathcal{C} = \{(2, 1)\}$. These conflict, $Z = \{(2, 2)\}$, $H = \{(1, 1)\}$, and deleting either member costs $1 \le |H|$.

The following fact is used in Step 9 of the proof. For finite index sets $\mathcal{I}, \mathcal{J}$ and $\Omega \subseteq \mathcal{I} \times \mathcal{J}$, a real matrix with rows indexed by $\mathcal{I}$ and columns by $\mathcal{J}$ is *supported on* Ω if its entries outside Ω are zero (entries in Ω may vanish).

Lemma 3.2 (Kernels at maximum support rank)**.** *Let $\Omega \subseteq \mathcal{I} \times \mathcal{J}$, where $\mathcal{I}$ and $\mathcal{J}$ are finite. Suppose that $\Theta : \mathbb{R}^{\mathcal{J}} \to \mathbb{R}^{\mathcal{I}}$ has maximum rank among all real matrices supported on Ω. Then, for every $\mathbf{x} \in \ker \Theta$, every $\mathbf{y} \in \ker \Theta^{\mathsf{T}}$, and every $(s, j) \in \Omega$, one has $\mathbf{x}_j \mathbf{y}_s = 0$.*

Proof. Suppose $\mathbf{x}_j \mathbf{y}_s \neq 0$. For real $\tau \neq 0$ the matrix $\Theta_\tau = \Theta + \tau e_s e_j^{\mathsf{T}}$ is supported on Ω, and $\Theta_\tau \mathbf{x} = \tau \mathbf{x}_j e_s$, so $e_s \in \operatorname{im} \Theta_\tau$. For every $\mathbf{w} \in \mathbb{R}^{\mathcal{J}}$, $\Theta \mathbf{w} = \Theta_\tau \mathbf{w} - \tau \mathbf{w}_j e_s \in \operatorname{im} \Theta_\tau$, so $\operatorname{im} \Theta + \mathbb{R} e_s \subseteq \operatorname{im} \Theta_\tau$. But $\mathbf{y}$ annihilates $\operatorname{im} \Theta$ (as $\mathbf{y}^{\mathsf{T}} \Theta = 0$) and $\mathbf{y}_s \neq 0$, so $e_s \notin \operatorname{im} \Theta$. Therefore $\operatorname{rank} \Theta_\tau > \operatorname{rank} \Theta$, contradicting maximality. □

Proof of Lemma 3.1. Plan. Steps 1–3 build a bipartite graph Γ and show that a small vertex cover of Γ yields the required deletion. Steps 4–10 prove that every matching of Γ has at most $|Z| + |H|$ edges, by encoding matchings as matrix ranks. Step 11 converts a maximum matching into a cover of the same size.

If there are no conflicts, retain everything; then $Z = H = \varnothing$ and the claim is immediate. Below, vector spaces carry their coordinate bases, and an empty minor has determinant 1, so the rank arguments remain valid for empty matchings or index sets.

Step 1 (the graph Γ). Let Z_L, Z_R be disjoint copies of Z, also disjoint from $\mathcal{R}$ and $\mathcal{C}$; write z_L, z_R for the copies of z. The bipartite graph Γ has sides

$$\mathcal{V}_L = \mathcal{R} \sqcup Z_L, \qquad \mathcal{V}_R = \mathcal{C} \sqcup Z_R,$$

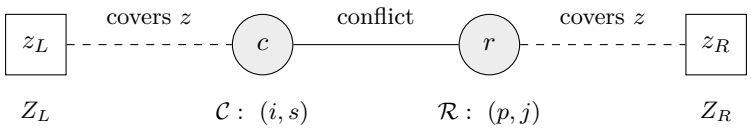


Figure 1: A conflict between $r = (p, j)$ and $c = (i, s)$ forces this path, where $z = (i, j) \in Z$. Different conflicts can share vertices and edges.

and three kinds of edges: (a) rc for conflicting $r \in \mathcal{R}$, $c \in \mathcal{C}$; (b) rz_R whenever $r \in \mathcal{R}$ covers $z \in Z$; (c) $z_L c$ whenever $c \in \mathcal{C}$ covers $z \in Z$. There are no edges between Z_L and Z_R. See Figure 1.

Step 2 (conflict-free members are isolated). Let $(p, j) \in \mathcal{R}$ cover $(i, j) \in Z$, so $p \to i$. A conflict witnessing $(i, j) \in Z$ supplies $(i, s) \in \mathcal{C}$ with $s \to j$, and this member conflicts with (p, j). Symmetrically, if $(i, s) \in \mathcal{C}$ covers $(i, j) \in Z$, so $s \to j$, a witness $(p, j) \in \mathcal{R}$ with $p \to i$ conflicts with it. Hence a member with no conflict has no edge of type (a), (b) or (c).

Step 3 (a small cover suffices). Let K be a vertex cover of Γ with $|K| \le |Z| + |H|$ and containing no isolated vertex. Put $\mathcal{R}' = \mathcal{R} \setminus K$, $\mathcal{C}' = \mathcal{C} \setminus K$. No conflict survives, since a conflict is an edge and K contains one of its ends. Conflict-free members are retained by Step 2. Let q be the number of points of Z covered by no member of $\mathcal{R}' \cup \mathcal{C}'$. Each other point z is covered by some $r \in \mathcal{R}'$ or $c \in \mathcal{C}'$; the edge rz_R (resp. $z_L c$) and $r \notin K$ (resp. $c \notin K$) force $z_R \in K$ (resp. $z_L \in K$). Distinct points give distinct copies, so $|K \cap (Z_L \sqcup Z_R)| \ge |Z| - q$ and

$$|\mathcal{R} \setminus \mathcal{R}'| + |\mathcal{C} \setminus \mathcal{C}'| = |K \cap (\mathcal{R} \sqcup \mathcal{C})| \le |K| - (|Z| - q) \le |H| + q, \tag{13}$$

which is (12). It remains to produce K.

Step 4 (a special maximum matching). Among maximum matchings of Γ choose $\mathcal{M}$ with the fewest edges of type (a); let k be their number. Let α and β be the sets of edges of $\mathcal{M}$ of types (b) and (c), and

$$\ell = |\alpha|, \qquad t = |\beta|, \qquad |\mathcal{M}| = k + \ell + t.$$

Let Γ_α, Γ_β be the subgraphs of Γ induced on $\mathcal{R} \sqcup Z_R$ and on $Z_L \sqcup \mathcal{C}$, and let $R_\alpha \subseteq \mathcal{R}$, $C_\beta \subseteq \mathcal{C}$ be the members matched by α, β.

Claim: α is a maximum matching of Γ_α, and β of Γ_β. If Γ_α had a larger matching, its symmetric difference with α would contain an α-augmenting path: a path in Γ_α whose edges alternate outside and inside α, with ends $r_0 \in \mathcal{R}$ and $z_R \in Z_R$ unmatched by α and interior vertices matched by α. All neighbors of z_R lie in $\mathcal{R}$, so z_R is unmatched in $\mathcal{M}$; and path edges outside α are outside $\mathcal{M}$, since they have type (b). If r_0 is unmatched in $\mathcal{M}$, the path augments $\mathcal{M}$, contradicting maximality. Otherwise the $\mathcal{M}$-edge e at r_0 has type (a); then the path augments $\mathcal{M} \setminus \{e\}$, giving a matching of size $|\mathcal{M}|$ with $k - 1$ edges of type (a), contradicting the choice of $\mathcal{M}$. The argument for β is the same with sides exchanged.

Step 5 (four matrices with $NL = AB$). For each pair with $x \to y$ take real variables a_{xy} and b_{xy}; set $a_{xy} = b_{xy} = 0$ when $x \not\to y$. For any real assignment define linear maps

$$\begin{array}{ccc} \mathbb{R}^{\mathcal{R}} & \xrightarrow{L} & \mathbb{R}^{Z} \\ {\scriptstyle B}\downarrow & & \downarrow{\scriptstyle N} \\ \mathbb{R}^{H} & \xrightarrow{A} & \mathbb{R}^{\mathcal{C}} \end{array}$$

by the matrix entries (row index first)

$$L_{(i,j),(p,j)} = a_{pi}, \quad N_{(i,s),(i,j)} = b_{sj}, \quad B_{(p,s),(p,j)} = b_{sj}, \quad A_{(i,s),(p,s)} = a_{pi}, \tag{14}$$

indices ranging over the indicated sets and every entry not of a listed form being zero. So L, A move the first coordinate from p to i, and N, B move the second coordinate from j to s.

Claim:

$$NL = AB. \tag{15}$$

Fix a column $(p, j) \in \mathcal{R}$ and a row $(i, s) \in \mathcal{C}$. In NL the only possible intermediate index is (i, j), and in AB it is (p, s); in each case the product of the two entries is $a_{pi}b_{sj}$ if the intermediate index exists, and the entry is 0 otherwise. If (p, j) and (i, s) conflict, then $(i, j) \in Z$ and $(p, s) \in H$, so both entries equal $a_{pi}b_{sj}$. If not, then $p \not\to i$ or $s \not\to j$, so $a_{pi}b_{sj} = 0$ and both entries vanish.

Step 6 (three rank conditions suffice). Suppose an assignment satisfies

$$\operatorname{rank} L = \ell, \qquad \operatorname{rank} N = t, \qquad \dim\ker A \geq k.$$

Then, by rank–nullity for A, $\operatorname{im}(AB) \subseteq \operatorname{im} A$, and (15),

$$|H| = \dim\ker A + \operatorname{rank} A \geq k + \operatorname{rank}(AB) = k + \operatorname{rank}(NL) \geq k + \ell + t - |Z|, \tag{16}$$

the last step because $\operatorname{rank}(NL) = \dim\operatorname{im} L - \dim(\operatorname{im} L \cap \ker N) \geq \ell - (|Z| - t)$. Thus $|\mathcal{M}| \leq |Z| + |H|$. Steps 7–10 produce such an assignment.

Step 7 (choice of coefficients). We exhibit finitely many nonzero polynomials in the variables a_{xy}, b_{xy} whose simultaneous nonvanishing gives what is needed. Facts (a), (b) are instances of the correspondence between generic matrix rank and maximum matching size [9, Section 5, Theorem 1]; we verify them directly.

(a) The matrix L. The entry of L at row $z = (i, j') \in Z$ and column $r = (p, j) \in \mathcal{R}$ can be nonzero only if $j' = j$ and $p \to i$, that is, only if rz_R is an edge of Γ_α. A nonzero square minor of L has a nonzero term in its permutation expansion, which is a matching of Γ_α between its columns and rows. Since α is maximum (Step 4), $\operatorname{rank} L \leq \ell$ for every assignment. Now let L_α be the $\ell \times \ell$ submatrix with columns R_α and rows the Z-endpoints of α. Group rows and columns by the second coordinate j. Entries between different groups vanish, and α matches within groups, so L_α is block diagonal with square blocks. Inside a block, the entry at row (i, j), column (p, j) is the variable a_{pi} or 0, and different positions carry different variables. Hence different permutations contribute different monomials to the block determinant and cannot cancel; the permutation given by α contributes a nonzero monomial. So each block determinant, and therefore $\det L_\alpha$, is a nonzero polynomial (a variable may occur in several blocks, but a product of nonzero polynomials is nonzero). Wherever $\det L_\alpha \neq 0$: $\operatorname{rank} L = \ell$ and the columns of L indexed by R_α form a basis of $\operatorname{im} L$.

(b) The matrix N^{T}. Its entry at row $(i', j) \in Z$, column $(i, s) \in \mathcal{C}$ can be nonzero only if $i' = i$ and $s \to j$, that is, along an edge of Γ_β. The same argument, with blocks of fixed first coordinate i and distinct variables b_{sj} inside each block, gives a nonzero polynomial $\det(N^{\mathsf{T}})_\beta$. Wherever it is nonzero, $\operatorname{rank} N = \operatorname{rank} N^{\mathsf{T}} = t$ and the columns of N^{T} indexed by C_β form a basis of $\operatorname{im} N^{\mathsf{T}}$.

(c) Support ranks. For every pair with $p \to i$ define

$$J_p = \{j : (p, j) \in \mathcal{R}\}, \quad S_i = \{s : (i, s) \in \mathcal{C}\}, \quad \Theta^{p,i} = (b_{sj})_{s \in S_i,\, j \in J_p}, \quad \Omega^{p,i} = \{(s, j) \in S_i \times J_p : s \to j\}.$$

For every assignment $\Theta^{p,i}$ is supported on $\Omega^{p,i}$. Let ρ be the maximum rank of real matrices supported on $\Omega^{p,i}$, and choose such a matrix with a nonzero $\rho \times \rho$ minor (the empty minor if $\rho = 0$). The same minor of $\Theta^{p,i}$ is a polynomial $\pi^{p,i}$ in the b-variables. The entries of $\Theta^{p,i}$ at positions of $\Omega^{p,i}$ are distinct variables, so they can be assigned the entries of the chosen matrix; thus $\pi^{p,i}$ is a

nonzero polynomial. Wherever $\pi^{p,i} \neq 0$, $\Theta^{p,i}$ has rank ρ, the maximum among matrices supported on $\Omega^{p,i}$.

Simultaneous choice. The product of $\det L_\alpha$, $\det(N^{\mathsf{T}})_\beta$ and all $\pi^{p,i}$ is a nonzero real polynomial, hence nonzero at some real assignment (induction on the number of variables). Fix such an assignment from now on. It gives $\operatorname{rank} L = \ell$, $\operatorname{rank} N = t$, the two basis properties, and maximum support rank for every $\Theta^{p,i}$.

Step 8 (lifts). Let $R_0 \subseteq \mathcal{R}$ be the set of the k left endpoints of the type-(a) edges of $\mathcal{M}$. For each such edge rc ($r \in R_0$, $c \in \mathcal{C}$), the basis properties in (a), (b) give coefficients with

$$\mathbf{u}^r = e_r + \sum_{r' \in R_\alpha} \lambda^r_{r'} e_{r'} \in \ker L, \qquad \mathbf{v}^c = e_c + \sum_{c' \in C_\beta} \mu^c_{c'} e_{c'} \in \ker N^{\mathsf{T}}. \tag{17}$$

Since $\mathcal{M}$ is a matching, $R_0 \cap R_\alpha = \varnothing$ and $c \notin C_\beta$. Hence the r''-coordinate of $\mathbf{u}^r$ is 1 if $r'' = r$ and 0 otherwise ($r, r'' \in R_0$), and $\mathbf{v}^c_c = 1$.

Step 9 (kernel restriction). *Claim:* at the fixed assignment,

$$\mathbf{u}_{(p,j)}\mathbf{v}_{(i,s)} = 0 \quad \text{for every conflicting } (p,j) \in \mathcal{R},\ (i,s) \in \mathcal{C} \text{ and all } \mathbf{u} \in \ker B,\ \mathbf{v} \in \ker N^{\mathsf{T}}. \tag{18}$$

Fix such a conflict; then $p \to i$, $j \in J_p$, $s \in S_i$ and $(s,j) \in \Omega^{p,i}$. Put $\mathbf{x} = (\mathbf{u}_{(p,j')})_{j' \in J_p}$ and $\mathbf{y} = (\mathbf{v}_{(i,s')})_{s' \in S_i}$.

(i) $\Theta^{p,i}\mathbf{x} = 0$. Take a row $s' \in S_i$. If $s' \to j'$ for some $j' \in J_p$, then (p,j') and (i,s') conflict, so $(p,s') \in H$. Row (p,s') of B is supported on the columns (p,j''), $j'' \in J_p$, so the (p,s')-coordinate of $B\mathbf{u} = 0$ reads

$$\sum_{j'' \in J_p} b_{s'j''} \mathbf{u}_{(p,j'')} = 0. \tag{19}$$

If there is no such j', all coefficients in (19) vanish and it holds trivially, whether or not $(p,s') \in H$.

(ii) $(\Theta^{p,i})^{\mathsf{T}}\mathbf{y} = 0$. Take a column $j' \in J_p$. If $s' \to j'$ for some $s' \in S_i$, the resulting conflict gives $(i,j') \in Z$. Column (i,j') of N is supported on the rows (i,s''), $s'' \in S_i$, so the (i,j')-coordinate of $N^{\mathsf{T}}\mathbf{v} = 0$ reads

$$\sum_{s'' \in S_i} b_{s''j'} \mathbf{v}_{(i,s'')} = 0. \tag{20}$$

Otherwise all coefficients vanish.

By Step 7(c), Lemma 3.2 applies to $\Theta^{p,i}$ with support $\Omega^{p,i}$ and gives $\mathbf{x}_j \mathbf{y}_s = 0$, which is (18). (The perturbation inside the proof of Lemma 3.2 is hypothetical and concerns only $\Theta^{p,i}$; the assignment is not changed.)

Step 10 ($\dim\ker A \geq k$). Let rc be a type-(a) edge of $\mathcal{M}$; then r, c conflict. Taking $\mathbf{v} = \mathbf{v}^c$ in (18) and using $\mathbf{v}^c_c = 1$: every $\mathbf{u} \in \ker B$ has $\mathbf{u}_r = 0$, for every $r \in R_0$. Now suppose $\sum_{r \in R_0} \kappa_r B\mathbf{u}^r = 0$. Then $\mathbf{u} = \sum_r \kappa_r \mathbf{u}^r \in \ker B$, so $\mathbf{u}_r = 0$ for $r \in R_0$; but $\mathbf{u}_r = \kappa_r$ by Step 8. So the k vectors $B\mathbf{u}^r$ are linearly independent, and $A(B\mathbf{u}^r) = NL\mathbf{u}^r = 0$ by (15) and (17). Hence $\dim\ker A \geq k$. With Step 7 this gives the three conditions of Step 6, so

$$|\mathcal{M}| = k + \ell + t \leq |Z| + |H|.$$

Step 11 (from the matching to a cover). We give the matching-to-cover argument of Kőnig's theorem [17]. Starting at all $\mathcal{M}$-unmatched vertices of $\mathcal{V}_L$, follow edges outside $\mathcal{M}$ from left to right and edges of $\mathcal{M}$ from right to left; let W_L, W_R be the sets of reached vertices on the two sides. No unmatched right vertex is reached, since an alternating path to it would augment $\mathcal{M}$.

The ends of an edge of $\mathcal{M}$ are both reached or both unreached: a reached right end leads to its mate, and a matched left vertex can only be reached through its mate. Put

$$K = (\mathcal{V}_L \setminus W_L) \cup W_R.$$

K covers every edge: an edge from W_L to $\mathcal{V}_R \setminus W_R$ can be neither outside $\mathcal{M}$ (it would be traversed) nor in $\mathcal{M}$ (same reachability of ends). K contains exactly one end of each edge of $\mathcal{M}$ and no unmatched vertex (unmatched left vertices lie in W_L; unmatched right vertices are not in W_R). So $|K| = |\mathcal{M}| \le |Z| + |H|$, and K contains no isolated vertex. Step 3 applied to K proves the lemma. □

4 The extremal contradiction

Proposition 4.1. *There is no nonempty finite oriented graph G in which every vertex has an in-neighbor and whose one-step image operator $F(U) = \{y \in V(G) : x \to y \text{ for some } x \in U\}$ satisfies*

$$|U| + |F^2(U)| < 2|F(U)| \qquad (\varnothing \neq U \subsetneq V(G)), \tag{21}$$

where $F^2 = F \circ F$.

Proof. Plan. Assume G exists. Choose an extremal compatible pair (P, Q) (Steps 1–2), build larger candidate families from complements of second images (Steps 3–5), and repair them with Lemma 3.1 at a cost that the objective absorbs (Steps 6–9).

Step 1 (coordinate images). Put $X = V(G)$. For $\Pi \subseteq X \times X$ let

$$F_1\Pi = \{(i,j) : (p,j) \in \Pi \text{ and } p \to i \text{ for some } p\}, \qquad F_2\Pi = \{(i,j) : (i,s) \in \Pi \text{ and } s \to j \text{ for some } s\},$$

and $F_1^2 = F_1 \circ F_1$, $F_2^2 = F_2 \circ F_2$. For fixed j, the column $\Pi_j = \{p : (p,j) \in \Pi\}$ satisfies $(F_1\Pi)_j = F(\Pi_j)$ and $(F_1^2\Pi)_j = F^2(\Pi_j)$; likewise F_2 acts as F on each row $\{s : (i,s) \in \Pi\}$. Let $\Delta = \{(v,v) : v \in X\}$. Call $P, Q \subseteq X \times X$ *compatible* if $F_1P \cap F_2Q = \varnothing$. By definition, $(i,j) \in F_1P \cap F_2Q$ if and only if there are $(p,j) \in P$ and $(i,s) \in Q$ with $p \to i$ and $s \to j$. So, taking the arc relation of G in Lemma 3.1 with left family P and right family Q: compatibility means there are no conflicts, and F_1P (resp. F_2Q) is the set of points covered by members of the left (resp. right) family.

Step 2 (extremal pair). The pair (Δ, Δ) is compatible: a point $(i,j) \in F_1\Delta \cap F_2\Delta$ would require $j \to i$ and $i \to j$. As X is finite, we may choose a compatible pair with $\Delta \subseteq P \cap Q$ maximizing $M + d$, where

$$M = |P| + |Q|, \qquad d = |X|^2 - |F_1P| - |F_2Q|.$$

By compatibility, d is the number of points outside $F_1P \cup F_2Q$.

Step 3 (columns of P and rows of Q are nonempty and proper). The column P_j contains j. Suppose $P_j = X$. Positive indegrees give $F(X) = X$, so $X \times \{j\} \subseteq F_1P$. Choose $w \to j$; then $(w,j) \in F_2\Delta \subseteq F_2Q$, contradicting compatibility. Likewise the row $Q_i = \{s : (i,s) \in Q\}$ contains i; if $Q_i = X$, then $\{i\} \times X \subseteq F_2Q$, and choosing $w \to i$ gives $(i,w) \in F_1\Delta \subseteq F_1P$, a contradiction.

Step 4 (summing the hypothesis). Apply (21) to $U = P_j$ and sum over $j \in X$, using Step 1; do the same for the rows of Q. Since $X \neq \varnothing$,

$$|P| + |F_1^2P| < 2|F_1P|, \qquad |Q| + |F_2^2Q| < 2|F_2Q|. \tag{22}$$

Step 5 (candidate families). Define

$$\mathcal{R} = (X \times X) \setminus F_2^2 Q, \qquad \mathcal{C} = (X \times X) \setminus F_1^2 P.$$

By (22) and the definitions of M and d,

$$|\mathcal{R}| + |\mathcal{C}| = 2|X|^2 - |F_2^2 Q| - |F_1^2 P| > 2|X|^2 + M - 2\big(|F_1 P| + |F_2 Q|\big) = M + 2d. \tag{23}$$

These families need not be compatible.

Step 6 ($\Delta \subseteq \mathcal{R} \cap \mathcal{C}$). If $(v, v) \in F_2^2 Q$, there is $s \to v$ with $(v, s) \in F_2 Q$; but $(s, s) \in P$ and $s \to v$ give $(v, s) \in F_1 P$, contradicting compatibility. If $(v, v) \in F_1^2 P$, there is $p \to v$ with $(p, v) \in F_1 P$; but $(p, p) \in Q$ gives $(p, v) \in F_2 Q$, the same contradiction.

Step 7 (diagonal members are conflict-free in $\mathcal{R}, \mathcal{C}$). A conflict between the left member (v, v) and a right member $(i, s) \in \mathcal{C}$ means $v \to i$ and $s \to v$. From $(s, s) \in P$, advancing the first coordinate along $s \to v \to i$ puts (i, s) in $F_1^2 P$, contrary to $(i, s) \in \mathcal{C}$. A conflict between a left member $(p, j) \in \mathcal{R}$ and the right member (v, v) means $p \to v$ and $v \to j$. From $(p, p) \in Q$, advancing the second coordinate along $p \to v \to j$ puts (p, j) in $F_2^2 Q$, contrary to $(p, j) \in \mathcal{R}$.

Step 8 ($|H| \le d$). Let H, Z be the sets attached to $\mathcal{R}, \mathcal{C}$ in Lemma 3.1. We claim

$$H \subseteq (X \times X) \setminus (F_1 P \cup F_2 Q), \qquad \text{hence} \qquad |H| \le d. \tag{24}$$

Fix a conflict $(p, j) \in \mathcal{R}$, $(i, s) \in \mathcal{C}$, $p \to i$, $s \to j$. If $(p, s) \in F_1 P$, choose $(u, s) \in P$ with $u \to p$; the path $u \to p \to i$ puts (i, s) in $F_1^2 P$, contrary to $(i, s) \in \mathcal{C}$. If $(p, s) \in F_2 Q$, choose $(p, s') \in Q$ with $s' \to s$; the path $s' \to s \to j$ puts (p, j) in $F_2^2 Q$, contrary to $(p, j) \in \mathcal{R}$.

Step 9 (prune and compare). Lemma 3.1 gives conflict-free $P' \subseteq \mathcal{R}$ and $Q' \subseteq \mathcal{C}$. By Steps 6–7 and its retention clause, $\Delta \subseteq P' \cap Q'$; by Step 1, $F_1 P' \cap F_2 Q' = \varnothing$. So (P', Q') competes in Step 2. Put

$$d' = |X|^2 - |F_1 P'| - |F_2 Q'|,$$

the number of points outside $F_1 P' \cup F_2 Q'$. By Step 1 these are exactly the points covered by no member of $P' \cup Q'$, so at most d' points of Z are uncovered after pruning. By (12) and (24), at most $d + d'$ members were deleted. With (23),

$$|P'| + |Q'| + d' \ge |\mathcal{R}| + |\mathcal{C}| - (d + d') + d' > M + 2d - (d + d') + d' = M + d,$$

contradicting the maximality of $M + d$. □

Proof of Theorem 1.1. If the theorem were false, Proposition 2.1 would produce a nonempty finite oriented graph with positive indegrees satisfying (21) for every nonempty proper vertex set. Proposition 4.1 rules out such a graph. □

5 Weighted consequences

Theorem 1.1 yields the established weighted forms through Seacrest's equivalences [19], which concern the corresponding assertions over all nonempty finite oriented graphs. They are not new here and are quoted, not reproved.

Corollary 5.1 (Weighted second-neighborhood forms)**.** *Let D be a nonempty finite oriented graph with arc set* $\mathrm{Arc}(D)$*. For a vertex weighting $\eta : V(D) \to [0, \infty)$, write $\eta(S) = \sum_{x \in S} \eta(x)$. Then the following hold.*

(i) *For every vertex weighting $\eta : V(D) \to [0,\infty)$, there is a vertex v such that $\eta(N_1^+(v)) \le \eta(N_2^+(v))$.*

(ii) *There is a vertex weighting $\eta : V(D) \to [0,\infty)$ with $\sum_{x\in V(D)} \eta(x) = 1$ such that $\eta(N_1^+(v)) \le \eta(N_2^+(v))$ for every $v \in V(D)$. The entries of this weighting need not all be positive.*

(iii) *For every arc weighting $\omega : \mathrm{Arc}(D) \to [0,\infty)$, set $\omega(vs) = 0$ when $vs \notin \mathrm{Arc}(D)$, and for $v, s \in V(D)$ put*

$$\gamma_v^\omega(s) = \max\Big(\{0\} \cup \{\omega(us) - \omega(vs) : u \in V(D),\ v \to u \to s\}\Big).$$

Then there is a vertex v such that

$$\sum_{s\in V(D)} \gamma_v^\omega(s) \ge \sum_{u\in N_1^+(v)} \omega(vu).$$

The maximum is zero if there is no indicated two-step path. Such paths are determined by the arcs of D: an arc of weight zero is still present.

Proof. (iii). Seacrest's Proposition 2 states that his arc-weighted Conjecture 4 is equivalent to the unweighted conjecture, so Theorem 1.1 gives (iii).

(i). Apply (iii) with $\omega(us) = \eta(s)$ on every arc $u \to s$. We claim $\gamma_v^\omega(s) = \eta(s)$ if $s \in N_2^+(v)$ and $\gamma_v^\omega(s) = 0$ otherwise. If there is no path $v \to u \to s$ (this includes $s = v$, by orientation), the maximum is 0. If $v \to s$, every eligible difference is $\eta(s) - \eta(s) = 0$. If $s \in N_2^+(v)$, then $\omega(vs) = 0$ and every eligible difference is $\eta(s) \ge 0$. So the two sides of (iii) are $\eta(N_2^+(v))$ and $\eta(N_1^+(v))$. This is the specialization noted immediately after Seacrest's Proposition 2.

(ii). Let Φ_D be the $V(D) \times V(D)$ matrix with entry 1 at (v,s) when $s \in N_2^+(v)$, entry -1 when $s \in N_1^+(v)$, and 0 otherwise; then $(\Phi_D\eta)_v = \eta(N_2^+(v)) - \eta(N_1^+(v))$. Let $\overleftarrow{D}$ be D with all arcs reversed. Then $\Phi_{\overleftarrow{D}} = \Phi_D^{\mathsf{T}}$: s is an out-neighbor of v in $\overleftarrow{D}$ iff $v \in N_1^+(s)$ in D, and s is an exact second out-neighbor of v in $\overleftarrow{D}$ iff $s \ne v$, $s \not\to v$ and $s \to u \to v$ for some u in D, i.e. iff $v \in N_2^+(s)$ in D. Seacrest's Theorem 1 applies Farkas' lemma to give either a nonzero $\eta \ge 0$ with $\Phi_D\eta \ge 0$, or a $\xi \ge 0$ with $\Phi_D^{\mathsf{T}}\xi < 0$ at every coordinate. The latter says $\xi(N_2^+(v)) < \xi(N_1^+(v))$ in $\overleftarrow{D}$ for every v, contradicting (i) for $\overleftarrow{D}$. Normalizing the former η to total weight one proves (ii). □

6 Directed-cycle consequences

In an oriented graph with no directed triangle, every exact second out-neighbor is a non-neighbor. The first assertion below is the oriented-graph form of Thomassé's non-neighbor statement recorded by Sullivan [21, Conjecture 6.12]; the consequence using both minimum degrees is noted in [21, Section 3]. The regular case gives the girth-four lower bound in the Behzad–Chartrand–Wall minimum-order conjecture [1], also recorded in [21, Section 4].

Corollary 6.1 (Non-neighbors and directed cycles)**.** *Let D be a nonempty finite oriented graph of order n. For $v \in V(D)$, write*

$$N_1^-(v) = \{u : u \to v\}, \qquad \deg^\pm(v) = |N_1^\pm(v)|, \qquad \delta^\pm(D) = \min_{x\in V(D)} \deg^\pm(x),$$

and define the non-neighbors of v, excluding v itself, by $\overline{N}(v) = V(D) \setminus \big(\{v\} \cup N_1^+(v) \cup N_1^-(v)\big)$. Then the following hold.

(i) *If D has no directed 3-cycle, then some vertex v satisfies* $\deg^+(v) \le |\overline{N}(v)|$ *and* $n - 1 \ge 2\deg^+(v) + \deg^-(v)$.

(ii) *If* $\delta^+(D) \ge n/3$ *and* $\delta^-(D) \ge n/3$*, then D contains a directed 3-cycle.*

(iii) *If* $r \ge 1$ *is an integer,* $\deg^+(v) = \deg^-(v) = r$ *for every vertex, and D has directed girth exactly four, then* $n \ge 3r + 1$.

Here directed girth is the length of a shortest directed cycle.

Proof. (i). Suppose D has no directed 3-cycle. Let $w \in N_2^+(v)$, with a path $v \to u \to w$. By definition $w \ne v$ and $w \notin N_1^+(v)$; and $w \notin N_1^-(v)$, since $w \to v$ would close a directed 3-cycle. Thus $N_2^+(v) \subseteq \overline{N}(v)$ for every v, and Theorem 1.1 gives a vertex with $\deg^+(v) \le |N_2^+(v)| \le |\overline{N}(v)|$. Since D is oriented, $N_1^+(v)$ and $N_1^-(v)$ are disjoint, so

$$n - 1 = \deg^+(v) + \deg^-(v) + |\overline{N}(v)| \ge 2\deg^+(v) + \deg^-(v).$$

(ii). If D had no directed 3-cycle, (i) would give $n - 1 \ge 2\delta^+(D) + \delta^-(D) \ge n$, a contradiction. (No rounding is involved, so the case $3 \mid n$ is included.)

(iii). Directed girth four excludes directed 3-cycles, and in the regular case (i) reads $n - 1 \ge 3r$. □

Verification checklist

No step was left unverified in this rewrite. The load-bearing steps, in suggested order of scrutiny:

1. **Cover accounting** (Lemma 3.1, Steps 2–3): that conflict-free members are isolated, and that a cover of size $\le |Z| + |H|$ yields (12). This is where "uncovered after deletion" enters.

2. $NL = AB$ **and the rank count** (Steps 5–6): identity (15) entry by entry, then (16).

3. **One assignment for all conditions** (Step 7): $\operatorname{rank} L \le \ell$ uses maximality of α in Γ_α (Step 4, which uses the minimal choice of k); the three families of polynomials are nonzero and are made nonzero simultaneously.

4. **Kernel restriction** (Step 9, with Lemma 3.2): equations (19) and (20) hold for *every* row and column of $\Theta^{p,i}$, including those whose index is absent from H or Z. Then Step 10 gives $\dim\ker A \ge k$.

5. **Extremal step** (Proposition 4.1): columns and rows are proper (Step 3), so (21) applies; diagonal members are conflict-free (Step 7); $|H| \le d$ (Step 8); the final arithmetic (Step 9).

6. **Arc deletion** (Lemma 2.2, Steps 5–10): inclusion (6), the gain and loss sets, and the exact increment (8).

7. **Blow-up** (Lemma 2.3, Steps 4–8): the two image counts, the passage to (11), and the separate case $S = V(D)$.

8. **External inputs.** Theorem 1.1 uses only standard linear algebra and augmenting paths. Section 5 additionally relies on the quoted statements of [19], which were not rechecked here.

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